Global News Big-Data Analysis: Enhancing Ministry Services via Sentiment AI

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***Abstract*—The Ministry of Culture, Sports and Tourism of Korea operates the *NOW* (now.go.kr) platform, which aggregates a massive dataset of over 2 million articles from both domestic and foreign media. However, the sheer volume of data limits the ability of policymakers to extract qualitative insights regarding Korea's global reputation. This project aims to enhance the platform by integrating Artificial Intelligence (AI) features for sentiment analysis and automated reporting. This paper details the end-to-end pipeline developed for this purpose, including a custom automated data acquisition system, a robust preprocessing framework, and a comparative analysis of Large Language Models (LLMs) for text summarization. Specifically, we focus on the 2025 APEC Summit in Korea as a case study, utilizing a fine-tuned MMBert model for sentiment classification. Furthermore, we propose a novel reputation scoring metric inspired by behavioral economics to better quantify public sentiment impact.**

***Index Terms*—Big Data, Sentiment Analysis, NLP, Automated Data Mining, APEC 2025, Behavioral Economics, MMBert.**

I. Introduction

THE digital era has led to an exponential increase in the volume of news media produced globally. For government agencies, monitoring this vast information landscape is critical for understanding national reputation and policy impact. The Korean Ministry of Culture, Sports and Tourism (MCST) addresses this through *NOW* (now.go.kr), a platform dedicated to aggregating global news regarding Korea.

Despite compiling a dataset exceeding 2 million articles, the platform currently relies heavily on simple keyword searches and frequency counts. This creates a "data rich, information poor" environment where policymakers struggle to discern qualitative shifts in global opinion, such as the distinction between factual reporting and critical editorializing.

The objective of this project is to transform this raw data into functional insights. We propose an AI-driven pipeline that automates data ingestion, summarizes complex geopolitical texts, and classifies sentiment using advanced Natural Language Processing (NLP) models. This report outlines the technical implementation of this pipeline, focusing on the architectural challenges of handling legacy web systems and the optimization of LLMs for cost-effective summarization.

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Prof. D. Kong is with the Graduate School of AI Policy & Strategy, GIST, and served as the primary mentor for this project.

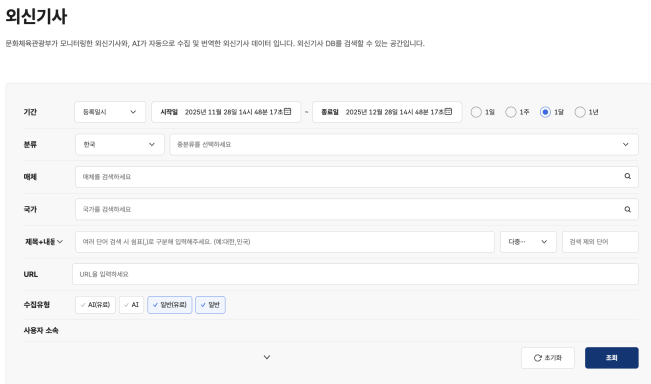

Fig. 1. The automated data ingestion pipeline. The script iterates through monthly intervals to bypass server query limits, aggregating the results into a unified dataset.

II. Data Acquisition and Preprocessing

A significant portion of the project involved architecting a solution to retrieve and clean legacy data from the *NOW* platform.

A. Automated Data Ingestion

The *NOW* platform does not provide a public API or a bulk download feature for its historical archive (2006–2025). Manual downloading was infeasible due to the scale of the dataset (2M+ articles).

To overcome this, I developed a custom automation script using Python and Selenium. The primary constraint imposed by the server was a limit on the number of records displayable per query. To bypass this, the script was designed to iterate through the database chronologically:

1) **Segmentation:** The time frame was divided into monthly intervals (e.g., 2006-01 to 2006-02).
2) **Traversal:** The script automatically navigated the DOM to select "Global Media" sections.
3) **Extraction:** Data was downloaded as discrete '.xlsx' files to avoid memory overflow and ensure checkpointing.

This approach ensured complete data coverage without triggering server-side blocking mechanisms.

B. Preprocessing and Standardization

The raw data exhibited significant inconsistencies typical of longitudinal datasets. I implemented rigorous cleaning using the *pandas* library:

- **Column Standardization:** Unified varying headers across different years into a standard schema.
- **Deduplication:** Removed redundant entries caused by re-published wire stories.
- **Entity Normalization:** Mapped varying country names and media outlets to standard ISO codes (e.g., "S. Korea", "ROK" → "KR").
- **Temporal Cleaning:** Converted diverse date formats into the standard `YYYY-MM-DD` format to facilitate time-series analysis.

This process resulted in a clean, fully usable dataset ready for model ingestion.

III. LLM Summarization Analysis

News articles regarding international relations are often lengthy and dense. To improve the efficiency of the sentiment analysis model (MMBert), it was necessary to first generate concise summaries that retained the semantic nuance of the original text (fact vs. opinion).

A. Comparative Experiment

I conducted an experiment to identify the optimal Large Language Model for this task, balancing computational cost against semantic accuracy. I focused on a subset of articles related to the "APEC 2025 Korea Summit."

I compared three models:

1) **Qwen (12B Parameters):** A lightweight open-source model running on local hardware.
2) **Gemma (12B Parameters):** A mid-sized open-weight model running on local hardware.
3) **Gemini 3.0 Pro:** SOTA model used as the "Ground Truth" baseline due to its high reasoning capabilities.

B. Evaluation Methodology

A random sample of 5 articles was processed by all models. The utilization of **ChatGPT-5 Pro** as an evaluator served a twofold purpose. First, it provided an objective comparison of the generated summaries against the original text and the Gemini 3.0 Pro baseline. Second, it acted as a necessary linguistic verification mechanism; as I am a non-native Korean speaker, the model was essential for auditing the summaries to ensure that complex Korean diplomatic nuances were accurately preserved without semantic drift. The evaluation criteria included: retention of key geopolitical entities, accurate reflection of sentiment tone, and conciseness (< 300 words).

IV. Case Study: APEC 2025 Analysis

To validate the efficacy of the proposed pipeline, we isolated data related to the APEC 2025 Summit held in the Republic of Korea. Filtering the dataset for the keyword "APEC" yielded approximately 6,000 relevant articles spanning the observation period (Oct 2024 – Nov 2025).

The summaries generated by the Gemma model were subsequently processed by a pre-trained **MMBert** (Multimodal BERT) model provided by my research team [3]. This model was fine-tuned to detect sentiment stance (Positive, Negative, Neutral), specifically within the context of diplomatic news coverage.

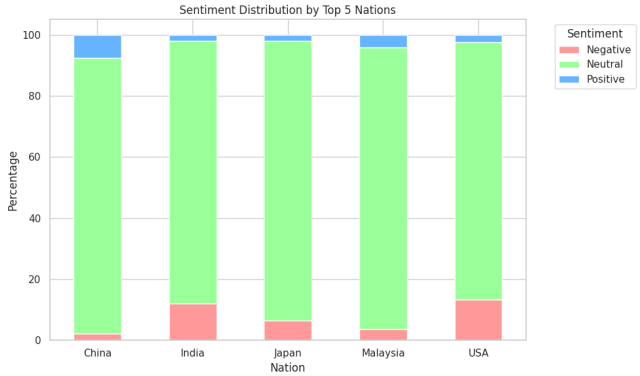

Fig. 2. Sentiment distribution by the top 5 reporting nations. A divergence is observable where Western allies (USA) and regional actors (India) show higher negative polarity compared to the neutral stance of China.

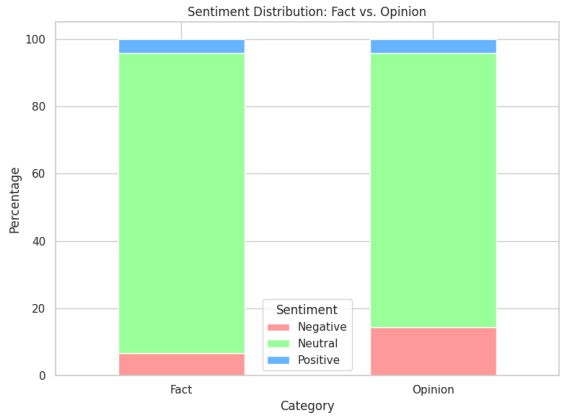

Fig. 3. Sentiment distribution categorized by article type. 'Opinion' pieces exhibit significantly higher polarity (Negative/Positive) compared to the predominantly 'Neutral' factual reporting.

A. Sentiment Divergence by Region

The analysis revealed distinct clusters of sentiment across different geopolitical actors. As illustrated in Fig. 2, coverage from **China** and **Malaysia** maintained an overwhelmingly neutral-to-positive stance. Conversely, coverage from the **USA** and **India** exhibited a notably higher proportion of critical (negative) sentiment, diverging from the expected diplomatic alignment.

B. Differentiation between Fact and Opinion

A critical finding was the model's capability to distinguish between "Fact" and "Opinion" pieces. As shown in Fig. 3, factual articles (reporting on logistics, schedules, and protocols) were overwhelmingly neutral. In contrast, opinion pieces, which often discussed economic fallout or controversial policy implications, contained the majority of the negative sentiment.

C. Temporal Trends

The temporal analysis (Fig. 4) indicates a marked increase in coverage volume as the summit approached. Notably, negative sentiment (red line) remained relatively stable and low,

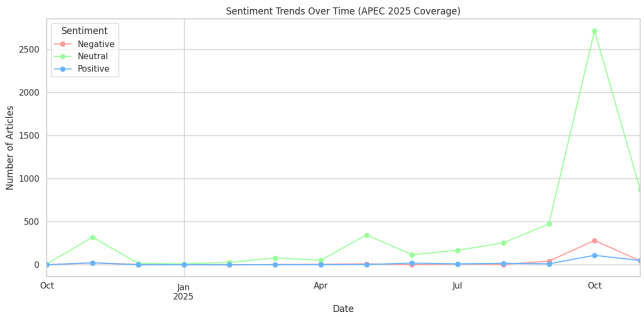

Fig. 4. Longitudinal sentiment trends from Oct 2024 to Nov 2025. The spike in 'Neutral' coverage correlates with the official summit dates.

whereas neutral coverage (green line) surged. This suggests that the event was largely successful in managing public perception and preventing widespread negative backlash.

V. Proposed Scoring System

To quantify the intensity and directionality of Korea's reputation, we developed a Sentiment-based Impact Score. Rather than treating positive and negative news symmetrically, this metric integrates the volume of coverage with weighted sentiment to reflect the actual influence on public opinion.

A. Weighted Impact Formula

Standard sentiment analysis often assigns equal weight to opposing sentiments (e.g., $+1$ vs. -1). However, drawing from *Prospect Theory* in behavioral economics, it is established that "losses loom larger than gains"—meaning negative news has a significantly stronger impact on opinion formation than positive news [2].

To reflect this asymmetry, we implemented the following scoring formula:

$$\text{Impact Score} = (1.0 \times N_{pos}) + (0.1 \times N_{neu}) + (-1.5 \times N_{neg}) \tag{1}$$

Where:

- **Positive (N_{pos}):** Weighted at **+1.0**, representing standard favorable coverage[cite: 136, 144].
- **Neutral (N_{neu}):** Weighted at **+0.1**, acknowledging that mere visibility provides a marginal positive effect.
- **Negative (N_{neg}):** Weighted at **-1.5**. Research suggests individuals perceive losses about 1.5 to 2.5 times more intensely than gains; we applied the conservative lower bound of 1.5 to ensure stability.

This "risk-aware" metric allows policymakers to identify not just the volume of news but also the severity of reputational risks, where a small number of negative articles can outweigh a larger volume of neutral reporting.

VI. Future Enhancements

To advance beyond static keyword monitoring, a future proposal is the implementation of a dedicated **Event Analysis Module**. This system is designed to track public opinion shifts across the lifecycle of major national events—such as the APEC 2025 Summit or political elections—by isolating sentiment data into three distinct temporal phases: *Pre-Event Expectations*, *Event Unfolding*, and *Post-Event Consequences*.

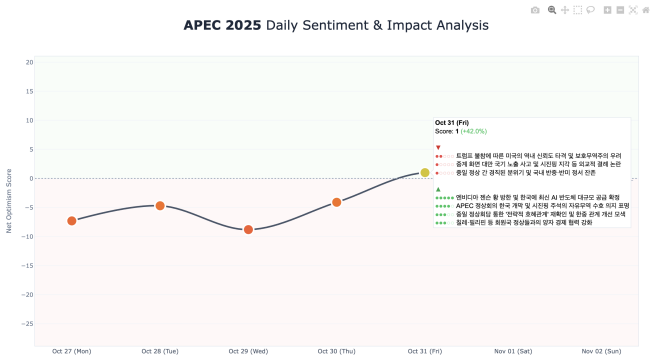

Fig. 5. Prototype of the Event Analysis Module applied to APEC 2025. The chart tracks the daily 'Net Optimism Score' throughout the summit week, identifying specific drivers of sentiment with an impact-based scoring.

A. Dynamic Event Tracking Pipeline

The proposed workflow utilizes a hybrid approach of human curation and LLM-driven automation:

- **Curated Event List:** A pre-defined list of high-impact events (e.g., APEC Summit Week, National Election) is maintained to ensure consistent monitoring of top-tier issues.
- **Custom Event Search:** Users can query specific incidents (e.g., "Coupang data breach"). An LLM normalizes the query, identifies the relevant date range, and hierarchically extracts keywords to filter the dataset.
- **Narrative Construction:** The model generates summaries for the week leading up to the event, the event period itself, and the aftermath. This highlights how expectations and subsequent realities diverge, offering policymakers a clear storyline of media discourse.

B. International Context Integration

To situate Korean events within broader regional dynamics, the module integrates cross-border sentiment analysis. It automatically generates comparative summaries of how key partner nations (United States, Japan, and China) perceived the same event, isolating divergences in diplomatic narratives.

VII. Conclusion

This project successfully automated the Ministry's news analysis, replacing manual keyword searches with an AI-driven pipeline. By using the Gemma model to summarize thousands of articles, we transformed the NOW platform into a source of immediate geopolitical insight. The APEC 2025 case study validated this approach, revealing a sharp contrast between the critical coverage from Western allies and the neutral stance of regional economic partners.

We also introduced a "Sentiment-based Impact Score" to better measure reputational risk. Because negative news

causes more damage than positive news repairs, this metric weights negative articles more heavily, giving policymakers a realistic view of public perception. Finally, the proposed Event Analysis Module offers a clear roadmap for the future. By tracking sentiment trends before, during, and after major national events, the Ministry can shift from reactive monitoring to proactive strategy.

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